Homework 4

Sammy Abdella, sn3136

1.

1(a) The negative of an even integer is even:

Proof. Let's say n is an even integer. Then there's an integer k where $n = 2k$. Taking the negative, we have $-n = -2k = 2(-k)$. Since $-k$ is an integer, $-n$ is of the form 2(integer), so $-n$ is even □

1(b) The sum of any three consecutive integers is divisible by 3:

Proof. Let's say we have the three consecutive n, $n+1$, and $n+2$ for some integer n. Their sum is

$$n + (n+1) + (n+2) = 3n + 3 = 3(n+1).$$

since $n+1$ is an integer, the sum is a multiple of 3, so it's divisible by 3. □

2.

Use proof by contradiction to show that: There exist no integers a and b for which $18a + 6b = 1$:

Proof. If there's integers a, b such that $18a + 6b = 1$. Factor out 6:

$$18a + 6b = 6(3a + b).$$

So the left-hand side is divisible by 6. But the equation says $18a + 6b = 1$, and 1 is not divisible by 6, it's a contradiction. Therefore, there's no integers a and b such that $18a + 6b = 1$. □

3.

Let x be a real number. Let $\lfloor x \rfloor$ be the greatest integer $\leq x$ and $\lceil x \rceil$ be the smallest integer $\geq x$. Prove that

$$\lfloor x \rfloor = \lceil x \rceil \quad \text{if and only if} \quad x \text{ is an integer.}$$

Proof. ($\Rightarrow$) Assume $\lfloor x \rfloor = \lceil x \rceil$. Let this common value be m, so $m = \lfloor x \rfloor = \lceil x \rceil$. By definition of floor and ceiling,

$$m = \lfloor x \rfloor \leq x \leq \lceil x \rceil = m.$$

Therefore $x = m$. since m is an integer, x is an integer.

($\Leftarrow$) Assume x is an integer. Then x itself is the greatest integer $\leq x$ and also the smallest integer $\geq x$. Hence $\lfloor x \rfloor = x$ and $\lceil x \rceil = x$, so $\lfloor x \rfloor = \lceil x \rceil$. □

4.

Prove that for any integers a, b, if a is even or b is even then the product ab is even.

Proof. Let $a, b \in \mathbb{Z}$ and assume (a is even) $\vee$ (b is even). We prove ab is even by cases.

Case 1: a is even. Then $a = 2k$ for some integer k, so $ab = (2k)b = 2(kb)$, which is even.

Case 2: b is even. Then $b = 2\ell$ for some integer ℓ, so $ab = a(2\ell) = 2(a\ell)$, which is even.

in either case, ab is even. □

5.

Let $n \in \mathbb{Z}$. Use proof by contrapositive to prove the following statement: If $8 \nmid n^3$, then n is odd.

Proof. We prove the contrapositive: *If n is even, then $8 \mid n^3$.*

let's assume n is even. Then $n = 2k$ for some integer k. Cubing gives $n^3 = (2k)^3 = 8k^3$. since k^3 is an an integer, n^3 is divisible by 8, i.e., $8 \mid n^3$. So the contrapositive is true, so the original statement is true. □

6. Proof by contrapositive

Use proof by contrapositive to prove: If $\log_2(3)$ is rational, then there exist positive integers m and n such that $2^m = 3^n$.

Proof. let's assume $\log_2(3)$ is rational. Then there exist integers m and n with $n > 0$ such that

$$\log_2(3) = \frac{m}{n}.$$

so by the definition of logarithm, this means

$$2^{m/n} = 3.$$

raising both sides to the nth power gives me

$$2^m = 3^n.$$

Because $3 > 1$, we have $\log_2(3) > 0$, so $m/n > 0$ so $m > 0$. also $n > 0$ by assumption. Therefore there exist positive integers m, n such that $2^m = 3^n$. □